

I-Wei (Claire) Chen

Location: Phoenix, AZ **Phone:** (858) 648-2560 **Email:** i4chen@ucsd.edu
LinkedIn: <https://www.linkedin.com/in/i-wei-claire-chen-0348b2208/>

SUMMARY

Software engineer with hands-on experience developing and deploying wireless communication architectures, simulating and maintaining end-to-end system, familiar with C++ / Python / MATLAB and has 2 years of experience.

SKILLS

Certificate: Qualcomm 5G Training Certificate from Coursera

Programming Skills & Tools: C++ / C, MATLAB, Python, Shell scripting, Swift, HTML, JAVA, Simulink, Visual C#

Domain Knowledge: 4G, 5G, Wi-Fi, Linux, Version control tools

EDUCATION

MS in Electrical and Computer Engineering, University of California, San Diego San Diego, CA

- Focused on **Communication Theory & Systems** Sep. 2021 – Mar. 2023
- *Relevant Courses: Modern Communication Networks, Digital Communication, Information Theory, Communication Circuits Design*

BS in Electrical and Computer Engineering, National Chiao Tung University Hsinchu, Taiwan

- **Last 60 GPA: 4.01/4.3** Sep. 2016 – Jun. 2020
- **Academic Achievement Award** - Top 5% of class in 2019
- *Relevant Courses: Data Communications, Digital Signal Processing, Advanced Probability for Communication*

WORK EXPERIENCES

Senior System / Algorithm Engineer, Ripple Technology Milpitas, CA
(C++, MATLAB, Python, Git) May. 2023 – Present

- Implemented **802.11n / a / ac / ax** end-to-end system and time domain algorithm in **C++**.
- Maintained simulators performance, time domain and channel estimation algorithm.
- Simulated **HE SU / VHT / HT / Non-HT** performance of our simulator, analyzed test results, and evaluated efficiency, functions and features and corrected the test issues.
- Enhanced automation scripts in **Python** to support multiple options in design verification environment, resulting 20% reduction in post-processing time.
- Supported DV group for maintaining hardware regression performance, supported RTL bit matching.

Wireless System Engineer Intern, Marvell Technology Santa Clara, CA
(C++, Eclipse, Git, MATLAB) Jun. 2022 – Sep. 2022

- Simulated **5G Physical Uplink Control Channel (PUCCH)** based on 3GPP standards in C++.
- Found new thresholds for 16 antennas at Base Station receiver to determine whether the received signal is a DTX or ACK different PUCCH formats.
- Implemented DAS (distributed antenna system)-like environments for **5G link level simulator** to identify issues and enhance simulation efficiency.

PROJECT EXPERIENCES

GymBro 2023

- Built an iOS workout tracking app using Xcode, SwiftUI and Figma.

Wifi PHY Implementation - Modern Communication Network Course Oct. 2021

- Implemented a basic **802.11** modulation and demodulation, and packet processing system using **MATLAB**.
- Simulated the **OFDM modulation** system and estimate the channel and transmitted data.

DoA Estimation and Beamforming in 5G Communication System Feb. 2019 – Jun. 2019

- Used **direction of arrival (DoA)** and **beamforming** technique, constructing **Fourier**, **MVDR**, and **MUSIC** methods (algorithm) that are used in modern 5G systems to estimate the directions pointing to the information source.
- Implemented situations in transmitters and receivers to observe the performance of 5G systems in **MATLAB**.

Viterbi Decoder for Convolutional Coded System - Coding Theory Course Sep. 2019 – Nov. 2019

- Implemented the BPSK communication system over a binary symmetric channel and an AWGN channel using the **Viterbi decoding method** to compare system performance in different noise magnitudes using **C++**.